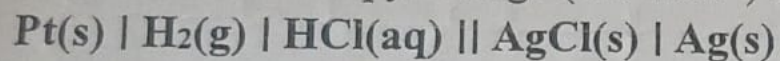


1. A current of 2 A is passed for 10 minutes through a solution of ZnSO_4 . The current efficiency is 85%. Calculate the amount of zinc (Zn) deposited at the cathode. (The molar mass of zinc is 65.38 g/mol). [4]

$$0.0105 / 0.69106$$

2. Calculate the entropy change ($\text{JK}^{-1}\text{mol}^{-1}$) for the following cell:



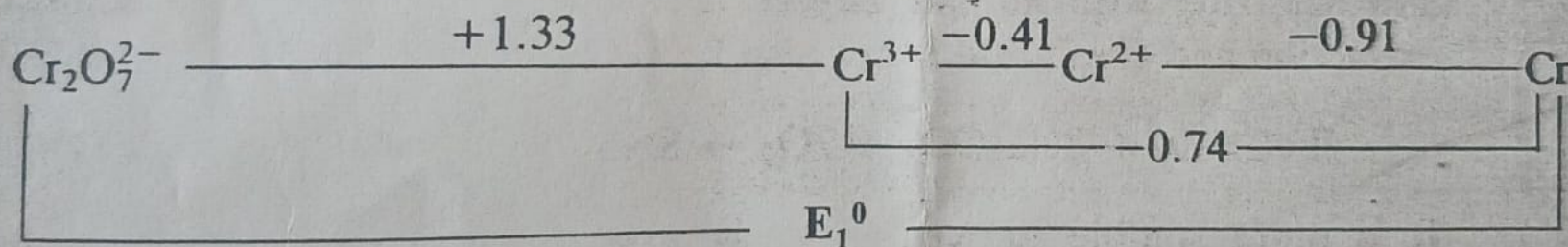
Given: $\text{EMF} = 0.080 \text{ V}$, Temperature coefficient of the cell $\text{EMF} = 4.2 \times 10^{-4} \text{ V/K}$, at 298 K. [4]

$$40.53$$

3. Calculate the cell potential for the cell $\text{Pb(s)} | \text{Pb}^{2+} (0.1 \text{ M}) || \text{Ni}^{2+} (0.05 \text{ M}) | \text{Ni(s)}$ at 298K. [$E_{\text{Pb}^{2+}/\text{Pb}}^0 = -0.126 \text{ V}$, $E_{\text{Ni}^{2+}/\text{Ni}}^0 = -0.257 \text{ V}$]. [4]

$$0.1221$$

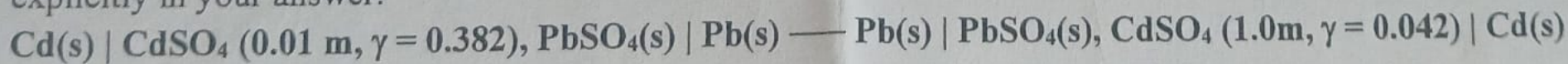
4. Calculate the value of E_1^0 from the following data. [4]



$$0.086$$

5. Draw the potentiometric titration curve for the titration of $\sim 0.01 \text{ N H}_2\text{SO}_4$ using $\sim 0.1 \text{ N NaOH}$ solution and explain the plot briefly. [4]

6. Write the cell reaction for the following cell and calculate the potential of the cell at 298K. Show all the steps explicitly in your answer. [10]



CHM204 MS1
Model Solution

S1. From Faraday's 2nd law of electrolysis; -

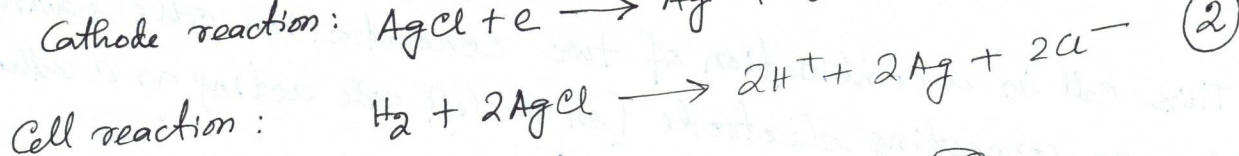
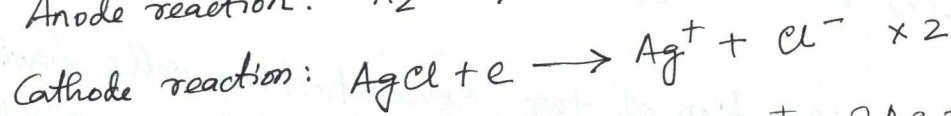
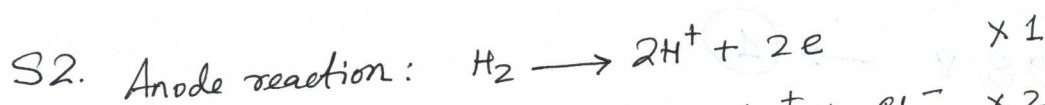
$$\boxed{W_{Zn} = Z_{Zn} \cdot C \cdot t} \quad \text{where } C = 2A, 85\% \text{ efficiency} \Rightarrow C = 1.7 \text{ Amp.}$$

$$t = 10 \text{ mins} = 600 \text{ s.}$$

$$Z_{Zn} = \frac{65.38}{2F} \text{ g coulomb}^{-1} \quad (1)$$

$$\therefore W_{Zn} = \frac{65.38}{2 \times 96500} \times \frac{1.7 \text{ amp} \times 600 \text{ s}}{\text{g coulomb}}$$

$$= 0.3455 \text{ g} \quad (2)$$

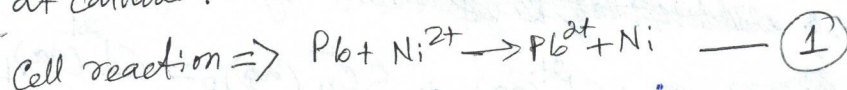
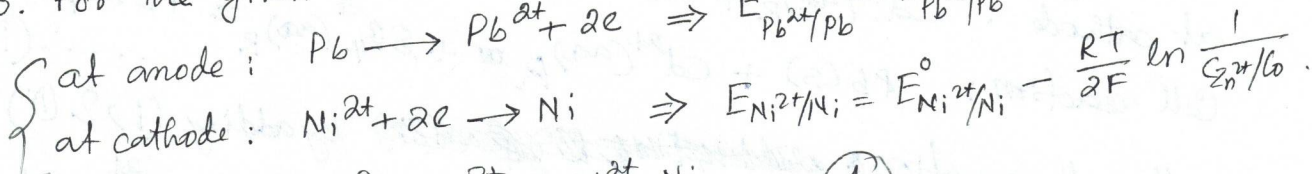
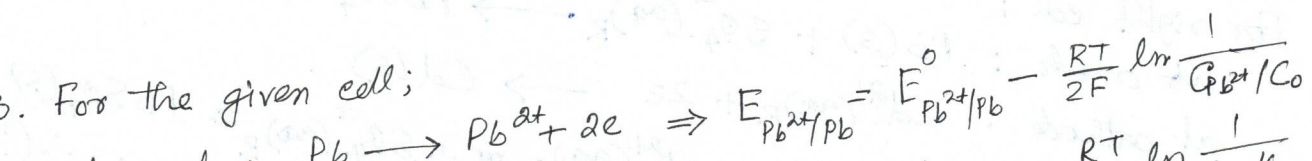


$$\therefore \Delta S = nF \left(\frac{\partial E}{\partial T} \right)_P \quad (1)$$

$$= 2 \times 96500 \text{ C mol}^{-1} \times 4.2 \times 10^{-4} \text{ V K}^{-1}$$

$$= 81.06 \text{ J K}^{-1} \text{ mol}^{-1} \quad (1)$$

S3. For the given cell;



$$E_{\text{cell}} = E_{Ni^{2+}/Ni} - E_{Pb^{2+}/Pb}$$

$$= (E_{Ni^{2+}/Ni}^0 - E_{Pb^{2+}/Pb}^0) - \frac{RT}{2F} \ln \frac{C_{Pb^{2+}}}{C_{Ni^{2+}}}$$



$n=6 \quad \Delta G$

$\Delta G = \Delta G_1 + \Delta G_2 + \Delta G_3 \rightarrow \textcircled{1} \quad \Delta G = -nFE^0$

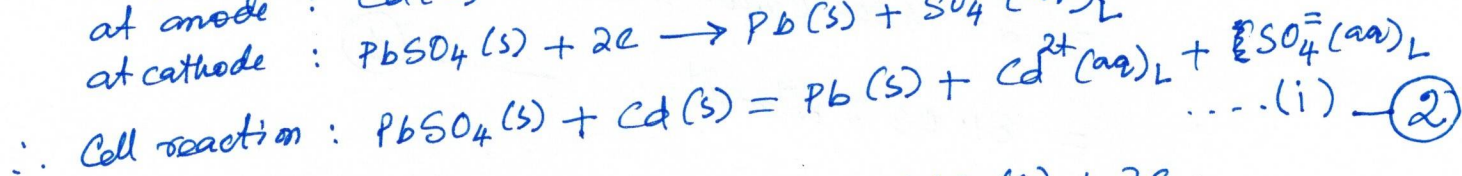
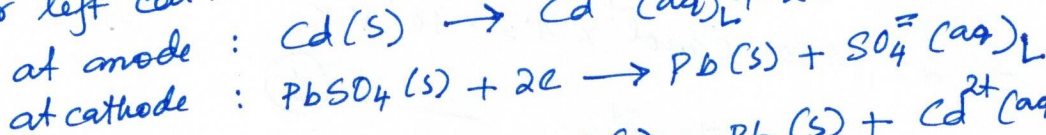
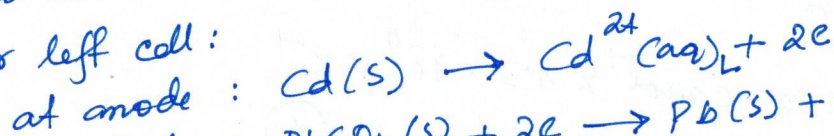
$-nFE^0 = -3 \times F \times 1.33 - 1 \times (-0.41) \times F - 2 \times F \times (-0.91)$

$-nE^0 = -1.79$

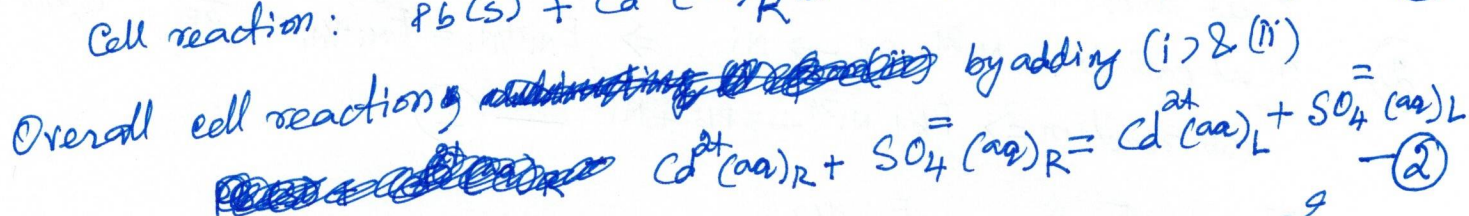
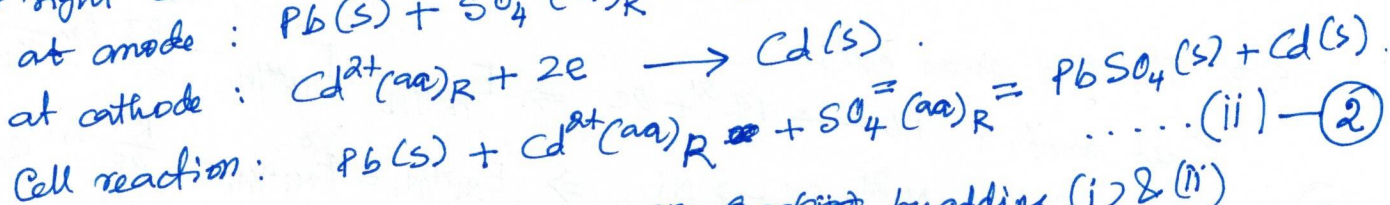
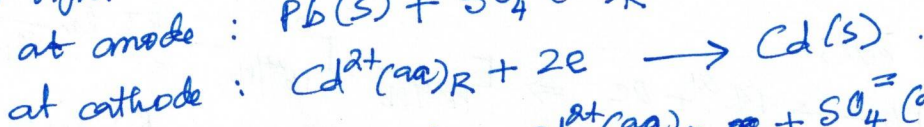
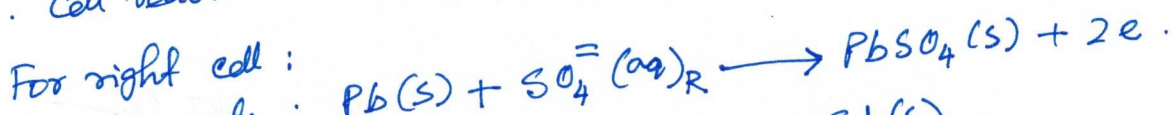
or $E^0 = 0.298 \text{ V} \rightarrow \textcircled{1}$

S5. This cell is a combination of two concentration cells having Pd(s) as connecting electrode (on the left side acting as a cathode and on the right side as an anode).

For left cell:

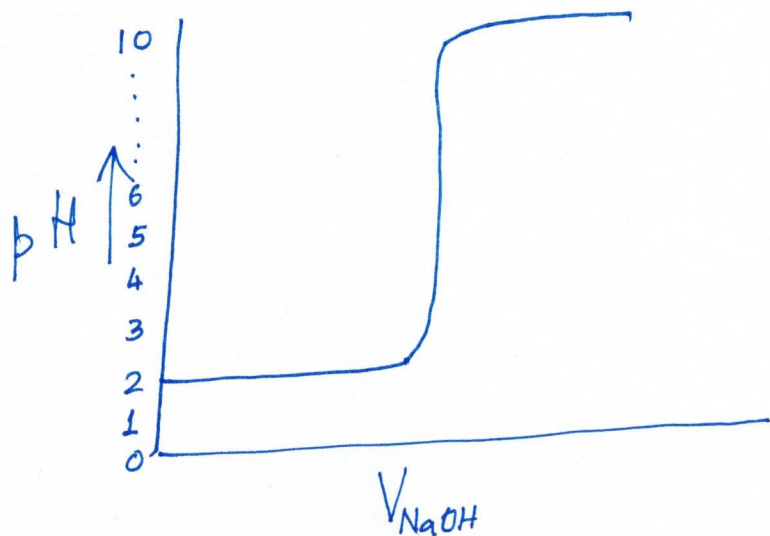


For right cell:



$\therefore E_{\text{cell}} = -\frac{RT}{2F} \ln \left\{ \frac{(a_{\text{Cd}^{2+}})_L \cdot (a_{\text{SO}_4^{2-}})_L}{(a_{\text{Cd}^{2+}})_R \cdot (a_{\text{SO}_4^{2-}})_R} \right\} \quad a = m \cdot \gamma^{\circ} \quad \textcircled{2}$

S5. In a potentiometric titration (acid-base), a solution of an acid ($\sim N/100$) is taken in a beaker and its pH is measured using a potentiometer. Then small portions of a solution of a base ($\sim N/10$) is added and pH is measured after each addition. This is continued till the pH reaches 9-10. The data of pH vs the volume of base added is plotted. For strong acids (HCl, H_2SO_4) ~~the is titrated~~ using a strong base ($NaOH/KOH$), the plot would look like the following.



As strong acids dissociates completely, the pH of the solution will not change significantly on adding strong base almost up to the neutralization point. At the neutralization point pH will suddenly jump from 2-3 to 9 within 1-3 drops of base solution ~~as~~ as shown in the figure.